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Mini Project Report

on

“REMOTE HEALTH MONITORING SYSTEMS”

Submitted in partial fulfillment of the requirements for the
First Semester of the Bachelor of Engineering Degree, towards the completion of the **Mini Project** under the **Innovation & Design Thinking Laboratory**,
Department of Basic Sciences.

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CERTIFICATE

This is to certify that the File Structures mini project entitled “Collision Resolution Using the Quadratic Probing Method” has been successfully carried out by Name1 (1CR25IS000), Name2 (1CR25IS000), and Name3 (1CR25IS000), bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract

The Remote Health Monitoring System is a hardware-based healthcare solution designed to continuously monitor patient vital parameters in real time using IoT technology. The system uses the ESP32 Development Board along with biomedical sensors such as MAX30102, DS18B20, and MPU6050 to measure heart rate, blood oxygen level (SpO₂), body temperature, and patient movement or fall detection. The collected sensor data is processed by the ESP32 and transmitted wirelessly to a monitoring dashboard for remote observation by healthcare staff and caregivers.

The system automatically detects abnormal health conditions and generates emergency alerts during critical situations, improving patient safety and reducing manual monitoring workload. The project provides a cost-effective, portable, and scalable solution suitable for hospitals, elderly care, and home healthcare monitoring. It also demonstrates the practical application of IoT, embedded systems, and real-time healthcare monitoring technologies in modern medical applications.

Keywords

- Remote Health Monitoring
- IoT Healthcare System
- ESP32 Development Board
- MAX30102 Sensor
- DS18B20 Temperature Sensor
- MPU6050 Sensor
- Heart Rate Monitoring
- SpO₂ Monitoring
- Fall Detection
- Real-Time Monitoring
- Embedded Systems
- Wireless Healthcare Monitoring
- Emergency Alert System
- Smart Healthcare
- Patient Monitoring System

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CHAPTER 1 – INTRODUCTION

Healthcare systems are increasingly adopting IoT-based technologies for continuous patient monitoring and emergency healthcare support. Traditional monitoring systems require manual observation and are mostly restricted to hospitals, making remote patient monitoring difficult. Delayed emergency response can lead to serious health complications, especially for elderly and critical patients.

The proposed Hardware-Based Remote Health Monitoring System continuously monitors patient health parameters such as heart rate, body temperature, and body movement using biomedical sensors and IoT technology. The system uses the ESP32 Development Board for data processing and wireless communication.

The hardware components used in the project are:

- HW-827 Pulse Sensor – Heart Rate Monitoring
- DS18B20 Sensor – Body Temperature Monitoring
- MPU6050 Sensor – Motion and Fall Detection
- ESP32 Development Board – Main Controller and Wi-Fi Communication

The collected sensor data is transmitted wirelessly to a monitoring dashboard, allowing caregivers and healthcare staff to monitor patients remotely in real time.

1.1 Evolution of Health Monitoring Systems

Earlier healthcare systems depended on manual patient observation and periodic medical checkups. With advancements in embedded systems and IoT technologies, modern healthcare monitoring systems now use smart sensors and wireless communication for continuous monitoring.

IoT-based healthcare systems improve:

- Real-time monitoring
- Emergency response
- Patient safety
- Remote healthcare accessibility

Affordable microcontrollers such as ESP32 and biomedical sensors have made portable health monitoring systems more practical and cost-effective.

1.2 Modern Remote Health Monitoring

Modern healthcare monitoring systems integrate sensors, embedded systems, wireless communication, and cloud/web technologies. These systems continuously collect patient data and automatically generate alerts during abnormal conditions.

The proposed system uses

- HW-827 sensor for pulse detection
- DS18B20 for temperature monitoring
- MPU6050 for fall detection
- ESP32 for real-time data transmission

The system supports:

- Continuous patient monitoring
- Wireless health data transmission
- Emergency alert generation
- Remote healthcare monitoring

CHAPTER 2 – PROBLEM STATEMENT

2.1 Description

Hospitals and caregivers face challenges in continuously monitoring multiple patients simultaneously. Manual monitoring may lead to delayed detection of abnormal conditions such as irregular heartbeat, high temperature, or accidental falls.

Existing systems are expensive, less portable, and often lack real-time remote monitoring capabilities.

The proposed hardware-based system addresses these problems by continuously monitoring patient health conditions using IoT sensors and wireless communication technologies.

2.2 Challenge Statement

To design and develop a low-cost hardware-based Remote Health Monitoring System using ESP32, HW-827 Pulse Sensor, DS18B20, and MPU6050 sensors for continuous patient monitoring, fall detection, wireless data transmission, and emergency alert generation

CHAPTER 3 – DESIGN THINKING PROCESS

3.1 Description of approach to solve the problem

a) Empathize

Discussions with caregivers and healthcare staff revealed difficulties in continuously monitoring elderly and critical patients. Delayed response during emergencies and lack of remote monitoring were major concerns.

Patients require continuous monitoring of:

- Heart rate
- Body temperature
- Body movement and falls

The proposed system addresses these challenges using IoT hardware sensors and wireless monitoring.

b) Define

The system should:

- Monitor patient vitals continuously
- Detect abnormal conditions automatically
- Detect patient falls using MPU6050
- Provide wireless data transmission
- Generate emergency alerts
- Reduce manual monitoring workload

c) Ideate

Different solutions such as wearable devices, mobile apps, and IoT dashboards were analyzed.

The final solution selected was a hardware-based IoT monitoring system using:

- ESP32
- HW-827 Pulse Sensor
- DS18B20
- MPU6050

d) Prototype

The hardware prototype consists of:

- ESP32 Development Board
- HW-827 Pulse Sensor
- DS18B20 Temperature Sensor
- MPU6050 Motion Sensor
- Breadboard and Jumper Connections

Working:

- HW-827 measures heart rate
- DS18B20 measures body temperature
- MPU6050 detects body movement and falls
- ESP32 processes and transmits data wirelessly

e) Test

The system was tested for:

- Heart rate monitoring accuracy
- Temperature measurement

- Fall detection performance
- Wi-Fi communication
- Real-time alert generation

The sensors successfully monitored patient health conditions and generated alerts during abnormal conditions.

3.2 METHODOLOGY

The Hardware-Based Remote Health Monitoring System continuously monitors patient health parameters using biomedical sensors connected to the ESP32 Development Board.

Step-by-Step Procedure

1. System Initialization

- ESP32 initializes all sensors and Wi-Fi connection.

2. Sensor Data Collection

- HW-827 measures heart rate.
- DS18B20 measures body temperature.
- MPU6050 detects motion and falls.

3. Data Processing

- ESP32 processes sensor readings.
- Values are compared with predefined threshold levels.

4. Wireless Transmission

- Sensor data is transmitted through Wi-Fi to the monitoring dashboard.

5. Patient Status Evaluation

- Stable

- Attention Needed
- Critical

6. Emergency Alert Generation

- Alerts are generated for:
 - Abnormal heart rate
 - High temperature
 - Fall detection

7. Continuous Monitoring

- The monitoring process repeats continuously in real time.

3.3 PROTOTYPE DESCRIPTION

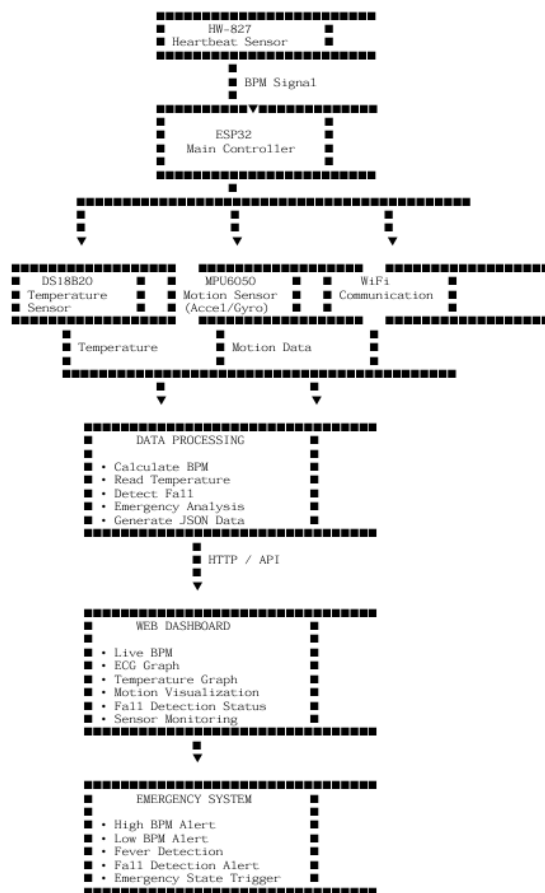
The Remote Health Monitoring System is a hardware-based IoT prototype designed for continuous patient monitoring.

Key Features

- Real-time heart rate monitoring using HW-827
- Body temperature monitoring using DS18B20
- Fall detection using MPU6050
- Wireless data transmission using ESP32
- Emergency alert generation
- Continuous healthcare monitoring

3.4 SYSTEM ARCHITECTURE

ESP32 IoT Health Monitoring System - Working Diagram



User Interface Layer

- Monitoring dashboard
- Emergency alert display
- Patient status visualization

Hardware Layer

- HW-827 Pulse Sensor
- DS18B20 Temperature Sensor
- MPU6050 Motion Sensor
- ESP32 Controller

Processing Layer

- Sensor data processing
- Threshold comparison
- Alert generation logic

Communication Layer

- Wi-Fi transmission using ESP32

CHAPTER 4 – IMPLEMENTATION

The implementation of the **Remote Health Monitoring System** involves integrating biomedical and safety monitoring sensors with the ESP32 Development Board to continuously monitor patient health parameters in real time. The system combines hardware components, embedded programming, wireless communication, and a monitoring dashboard to create a complete IoT-based healthcare solution.

The implementation process was carried out in multiple stages including hardware setup, sensor integration, data processing, wireless communication, dashboard monitoring, and alert generation.

4.1 Hardware Setup

The hardware components were assembled on a breadboard and connected to the ESP32 Development Board. The ESP32 acts as the central controller responsible for collecting, processing, and transmitting sensor data.

Hardware Components Used

Component	Function
ESP32 Development Board	Main controller and Wi-Fi communication
HW-827 Pulse Sensor	Heartbeat monitoring

DS18B20 Sensor	Body temperature monitoring
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MPU6050 Sensor	Motion and fall detection
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Breadboard & Jumper Wires	Circuit connections
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USB Power Supply	System power source
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The sensors were connected to the ESP32 using analog, I2C, and digital communication protocols. Proper voltage and grounding connections were ensured for stable operation.

4.2 Sensor Integration

HW-827 Pulse Sensor Integration

The HW-827 pulse sensor was connected to the ESP32 analog input pin to monitor the patient's heartbeat in real time.

The sensor detects pulse signals by measuring changes in blood flow and sends analog heartbeat data to the ESP32 for processing.

Purpose

- Monitor patient heartbeat rate
- Detect abnormal pulse conditions
- Provide continuous heart monitoring

DS18B20 Temperature Sensor Integration

The DS18B20 digital temperature sensor was connected to the ESP32 using a one-wire communication interface.

The sensor continuously measures body temperature and sends digital readings directly to the ESP32 controller.

Purpose

- Monitor patient body temperature
- Detect fever or abnormal temperature rise
- Improve healthcare observation accuracy

MPU6050 Sensor Integration

The MPU6050 sensor was connected using the I2C protocol to measure motion and orientation.

The sensor includes:

- Accelerometer
- Gyroscope

The MPU6050 continuously monitors patient movement and detects sudden falls or abnormal body motion.

Purpose

- Fall detection
- Movement monitoring
- Patient activity tracking

4.3 ESP32 Programming

The ESP32 Development Board was programmed using the **Arduino IDE** with Embedded C/C++ programming language.

Different sensor libraries were installed to simplify communication between ESP32 and the sensors.

Functions Implemented

- Sensor initialization
- Real-time data collection
- Threshold checking
- Wi-Fi communication
- Alert generation
- Dashboard data transmission

The ESP32 continuously reads sensor values and processes the information in real time.

4.4 Real-Time Data Processing

The collected sensor data is processed by the ESP32 microcontroller. The system continuously compares sensor readings with predefined threshold values

Example Threshold Conditions

Parameter	Threshold
Heart Rate	Too High / Too Low
Temperature	High Fever
Motion Detection	Sudden Fall

When sensor values exceed safe limits, the system identifies the patient condition as critical.

4.5 Wireless Communication

The ESP32 uses its built-in Wi-Fi module to transmit sensor data wirelessly to the healthcare monitoring dashboard.

Data Transmitted

- Heart Rate
- Body Temperature
- Patient Motion Status
- Emergency Alerts

Wireless communication enables remote patient monitoring from any location connected to the network.

4.6 Dashboard Monitoring System

A web-based monitoring dashboard was developed to display patient health data in real time.

Dashboard Features

- Real-time sensor data display
- Patient status monitoring
- Emergency alert notifications
- Graphical health parameter visualization
- Continuous data updates

The dashboard helps healthcare staff monitor patient conditions remotely and respond quickly during emergencies.

4.7 Emergency Alert System

The system includes an automatic emergency alert mechanism.

When abnormal conditions are detected:

- Visual alerts are displayed
- Warning notifications are generated
- Emergency monitoring is activated

Examples of emergency conditions include:

- High body temperature
- Abnormal heart rate
- Sudden patient fall

This feature improves patient safety and reduces emergency response time.

4.8 System Testing

The complete hardware system was tested under simulated healthcare conditions.

Testing Performed

- Heart rate monitoring accuracy
- Temperature sensor testing
- Fall detection testing
- Wi-Fi communication testing
- Dashboard update testing
- Emergency alert testing

The testing results confirmed stable operation, accurate monitoring, and successful real-time communication between hardware and dashboard systems.

4.9 Final Outcome

The implemented Remote Health Monitoring System successfully demonstrates:

- Real-time patient monitoring
- Wireless healthcare communication
- IoT-based emergency alert system
- Fall detection and movement monitoring
- Continuous health parameter tracking

The project provides a low-cost, scalable, and efficient healthcare monitoring solution suitable for hospitals, elderly care centers, and home healthcare applications.

CHAPTER 5 – RESULTS AND ANALYSIS

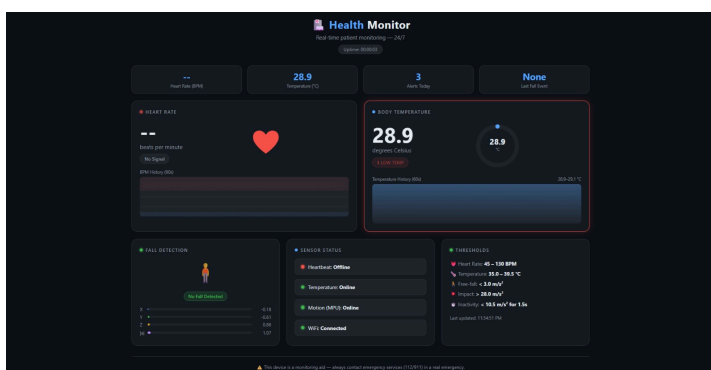
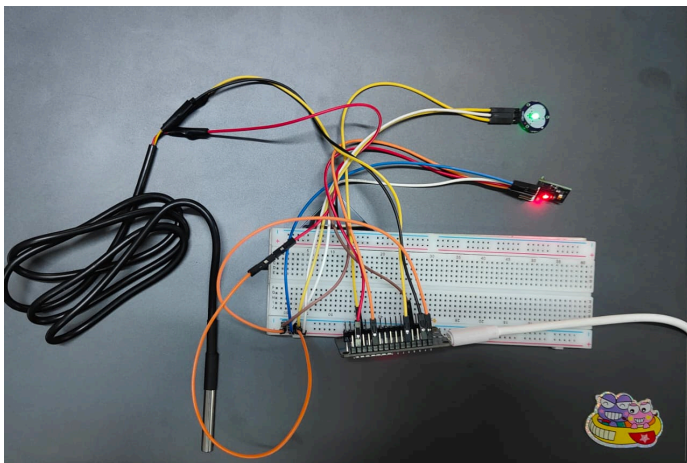
Hardware Results

The system successfully monitored:

- Heart rate
- Body temperature
- Patient movement

Results Obtained

- Accurate pulse detection using HW-827
- Stable temperature monitoring
- Successful fall detection using MPU6050
- Real-time wireless monitoring using ESP32
- Automatic emergency alert generation



CHAPTER 6 – CONCLUSION & FUTURE WORK

The Remote Health Monitoring System successfully demonstrates a low-cost and efficient healthcare monitoring solution using ESP32, HW-827 pulse sensor, DS18B20 temperature sensor, and MPU6050 motion sensor. The system continuously monitors patient health conditions in real time and sends the data wirelessly to the monitoring dashboard. It also generates emergency alerts during abnormal conditions such as high temperature, abnormal heartbeat, and fall detection.

The project improves patient safety, reduces manual monitoring workload, and demonstrates the practical use of IoT and embedded systems in healthcare applications.

Future Work

- Cloud storage integration for patient data
- Mobile application for remote monitoring
- AI-based health prediction system
- Addition of more biomedical sensors
- GPS and ambulance integration
- Wearable healthcare device development
- SMS and call alert system
- Advanced dashboard analytics and reports

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